



Google Prep Playlist — Senior DevOps / GCP / MLOps / AI Infra

For your background (5+ years GCP + infra), you should prepare like a:

- Senior Cloud Engineer
- Platform Engineer
- AI Infrastructure Engineer
- MLOps Engineer

This means:

- deep concepts
- troubleshooting
- scalability
- architecture
- tradeoffs
- production failures

NOT basic definitions.



PLAYLIST 1 — KUBERNETES (MOST IMPORTANT)

Core Architecture

Questions

1. Explain complete Kubernetes architecture in depth.
2. How does kube-apiserver work internally?
3. How does etcd maintain consistency?
4. What happens when you run:

```
kubectl apply -f app.yaml
```

5. Explain scheduler internals.

6. How does controller manager work?
 7. What are reconciliation loops?
 8. How does desired state management work?
 9. Difference between Deployment vs StatefulSet vs DaemonSet.
 10. How do ReplicaSets work internally?
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Networking

Questions

1. Explain Kubernetes networking from scratch.
 2. How does pod-to-pod communication work?
 3. What is CNI?
 4. Difference between ClusterIP, NodePort, LoadBalancer.
 5. Explain kube-proxy internals.
 6. iptables vs IPVS.
 7. Explain ingress architecture.
 8. NGINX ingress vs Gateway API.
 9. Debugging DNS failures inside cluster.
 10. How would you troubleshoot intermittent service communication?
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Storage

Questions

1. PV vs PVC vs StorageClass.
 2. Explain CSI drivers.
 3. Stateful workloads in Kubernetes.
 4. Dynamic provisioning.
 5. ReadWriteOnce vs ReadWriteMany.
 6. Kubernetes backup strategies.
 7. etcd backup/restore process.
 8. Persistent storage on GKE.
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Security

Questions

1. RBAC internals.
 2. ServiceAccount vs User.
 3. Network policies.
 4. Pod Security Standards.
 5. Secret management strategies.
 6. Workload Identity in GKE.
 7. Admission controllers.
 8. OPA/Gatekeeper usage.
 9. Runtime container security.
 10. Supply chain security in Kubernetes.
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Scaling + Reliability

Questions

1. HPA vs VPA vs Cluster Autoscaler.
 2. How does autoscaling work internally?
 3. Pod disruption budgets.
 4. Multi-region Kubernetes architecture.
 5. Zero downtime deployments.
 6. Blue-green vs Canary deployments.
 7. How do you design highly available clusters?
 8. Kubernetes failure domains.
 9. API server bottlenecks.
 10. Troubleshooting cluster instability.
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PLAYLIST 2 — GCP + GKE

GKE

Questions

1. Standard vs Autopilot GKE.
2. GKE architecture.

3. Node pools design strategies.
 4. Private GKE clusters.
 5. Workload Identity.
 6. GKE networking internals.
 7. Multi-cluster GKE architecture.
 8. GKE upgrades and maintenance windows.
 9. GPU workloads on GKE.
 10. Cost optimization strategies.
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Networking

Questions

1. Explain GCP VPC architecture.
 2. Shared VPC vs VPC peering.
 3. Cloud NAT internals.
 4. Load balancing types in GCP.
 5. Hybrid cloud connectivity.
 6. Cloud Armor usage.
 7. DNS architecture in GCP.
 8. Internal vs external load balancers.
 9. Troubleshooting network latency.
 10. Multi-region traffic routing.
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IAM + Security

Questions

1. GCP IAM hierarchy.
2. Custom roles vs predefined roles.
3. Organization policies.
4. Service accounts best practices.
5. Workload Identity Federation.
6. Security boundaries in GCP.
7. VPC Service Controls.
8. Secret Manager architecture.
9. KMS usage patterns.
10. Least privilege implementation.

Reliability + Operations

Questions

1. Designing HA systems on GCP.
 2. DR strategy for GKE.
 3. Multi-region architecture.
 4. Monitoring stack on GCP.
 5. Incident management.
 6. Cost optimization.
 7. Logging architecture.
 8. SLO/SLI/SLA design.
 9. Capacity planning.
 10. Production outage handling.
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PLAYLIST 3 — TERRAFORM + IaC

Terraform Fundamentals

Questions

1. Terraform lifecycle.
 2. State management internals.
 3. Remote state architecture.
 4. Terraform locking.
 5. Drift detection.
 6. Terraform graph theory.
 7. Modules design.
 8. Workspace strategy.
 9. Terraform performance optimization.
 10. Monorepo vs multirepo IaC.
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Enterprise IaC

Questions

1. Multi-environment design.
 2. Policy as code.
 3. Secret management.
 4. CI/CD for Terraform.
 5. Terragrunt usage.
 6. State corruption recovery.
 7. Handling large infra.
 8. Blue-green infra deployment.
 9. Infrastructure testing.
 10. Terraform security best practices.
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PLAYLIST 4 — PYTHON FOR PLATFORM ENGINEERS

Python Backend

Questions

1. Sync vs async Python.
 2. GIL explained.
 3. Threading vs multiprocessing.
 4. FastAPI architecture.
 5. REST API design.
 6. Dependency injection.
 7. Background workers.
 8. Python memory management.
 9. Python performance tuning.
 10. API rate limiting.
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Automation

Questions

1. Infrastructure automation using Python.
 2. Kubernetes automation.
 3. CI/CD automation.
 4. Writing production-grade scripts.
 5. Error handling strategies.
 6. Observability in Python apps.
 7. Retry patterns.
 8. Queue systems.
 9. Async APIs.
 10. API authentication methods.
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PLAYLIST 5 — MLOPS

ML Infrastructure

Questions

1. Complete ML lifecycle.
 2. Training vs inference infrastructure.
 3. Batch vs online inference.
 4. Feature stores.
 5. Model registry architecture.
 6. Experiment tracking.
 7. ML reproducibility.
 8. ML pipelines.
 9. Kubeflow architecture.
 10. MLflow internals.
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Production ML Systems

Questions

1. Deploying models on Kubernetes.
2. GPU scheduling.
3. Canary deployment for ML.
4. Model drift detection.

5. AI observability.
 6. Shadow deployment.
 7. A/B testing for ML.
 8. Scaling inference systems.
 9. Vector databases.
 10. Cost optimization for AI systems.
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PLAYLIST 6 — LLMOPS + AI AGENTS

RAG Systems

Questions

1. Explain RAG architecture.
 2. Embeddings explained deeply.
 3. Chunking strategies.
 4. Retrieval optimization.
 5. Hybrid search.
 6. Vector databases internals.
 7. Hallucination mitigation.
 8. Reranking systems.
 9. Context window optimization.
 10. Multi-agent RAG systems.
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AI Agents

Questions

1. What are AI agents?
2. Tool calling architecture.
3. Agent memory systems.
4. LangGraph architecture.
5. Multi-agent orchestration.
6. Agent failure handling.
7. AI workflow orchestration.
8. Human-in-the-loop systems.

9. Observability for agents.
 10. Security risks in AI agents.
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PLAYLIST 7 — SYSTEM DESIGN (VERY IMPORTANT)

Core System Design

Questions

1. Design YouTube.
 2. Design Uber.
 3. Design Kubernetes control plane.
 4. Design scalable CI/CD platform.
 5. Design distributed logging system.
 6. Design ML inference platform.
 7. Design vector search platform.
 8. Design global load balancing system.
 9. Design observability platform.
 10. Design ChatGPT-like system.
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Reliability

Questions

1. CAP theorem explained practically.
2. Consistency vs availability tradeoffs.
3. Circuit breaker patterns.
4. Retry patterns.
5. Distributed tracing.
6. Event-driven systems.
7. Queue architectures.
8. Rate limiting systems.
9. Caching strategies.
10. Disaster recovery systems.

PLAYLIST 8 — BEHAVIORAL (GOOGLE IMPORTANT)

Google heavily evaluates communication.

Leadership

Questions

1. Biggest production incident handled?
 2. Conflict with teammate?
 3. Major failure?
 4. Difficult migration?
 5. Mentoring junior engineers?
 6. Ownership example?
 7. Improving reliability?
 8. Handling ambiguity?
 9. Working under pressure?
 10. Tough stakeholder management?
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HOW TO USE THIS

DAILY METHOD

Daily

1 topic = 1 recording

Example:

- “Explain Kubernetes ingress deeply”
- 15 mins explanation
- upload privately to YouTube



BEST STRUCTURE FOR ANSWERS

Always structure answers:

- 1. Definition**
- 2. Why needed**
- 3. Architecture**
- 4. Workflow**
- 5. Real-world usage**
- 6. Tradeoffs**
- 7. Failure scenarios**
- 8. Debugging**
- 9. Scaling concerns**
- 10. Best practices**

This structure sounds VERY senior-level in interviews.



MOST IMPORTANT PLAYLISTS FOR YOU

Priority order:

1. Kubernetes
2. GKE
3. System Design
4. MLOps
5. LLMOps
6. Python backend
7. Behavioral

These align best with your:

- current experience
- future AI infra goals
- Google-level preparation.